PT INFINIX DELAPAN TEKNOLOGI NUSANTARA

T8 VOLGA I OPTICAL TRANSPORT PLATFORM

Official Carrier-Grade DWDM Technical Specifications & Architecture

1. PLATFORM OVERVIEW

The Volga I platform is a carrier-grade, highly scalable Dense Wavelength Division Multiplexing (DWDM) optical transport system engineered for long-haul backbones, regional rings, and high-density Data Center Interconnect (DCI) networks. Developed by global optical pioneer T8 Systems and distributed, adapted, and supported in Indonesia by PT Infinix Delapan Teknologi Nusantara (IDTN), Volga I provides up to 38.4 Tbps total transmission capacity per fiber pair across the C+L optical spectrum bands without electric regeneration.

2. CHASSIS FORM FACTORS & HARDWARE CONFIGURATIONS

- **Volga I Telecom (1U, 3U, 6U, 10U Modular Chassis)**
- **Volga I DCI (1U / 2U Compact High-Density)**
- **Volga I DCI V2 (6.4 Tbps per 2U Chassis)**

3. KEY OPTICAL & TRANSMISSION SPECIFICATIONS

Maximum Total Capacity	38.4 Tbps per fiber pair (C+L Band 96+96 channels)
Per-Channel Bitrate	100G, 200G, 400G, and 800G per wavelength
Reach without Regeneration	> 1,500 km Ultra Long-Haul (with integrated EDFA/Raman amplifiers)
Protection Switching	1+1 Optical Line Protection (OLP) with sub-50ms automatic failover
Optical ROADM	Colorless, Directionless, Contentionless, Flex-Grid (CDC-F ROADM)
Latency Profile	< 5 microseconds per km of fiber (deterministic transmission)
Physical Encryption	Wire-speed Layer 1 AES-256 hardware encryption compliant

4. FIBER HEALTH & NETWORK ORCHESTRATION

- **OTDR Monitoring System Ramenka**
- **NMS Titan & Fractal GUI Management**

PT INFINIX DELAPAN TEKNOLOGI NUSANTARA - OFFICIAL T8 SYSTEMS PARTNER

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